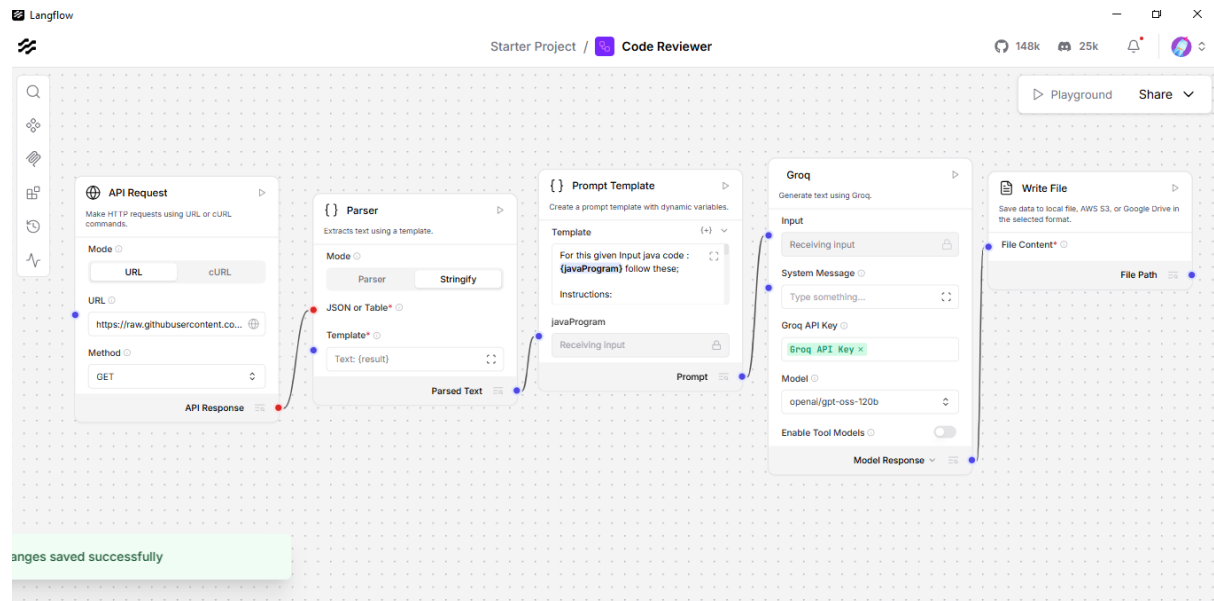
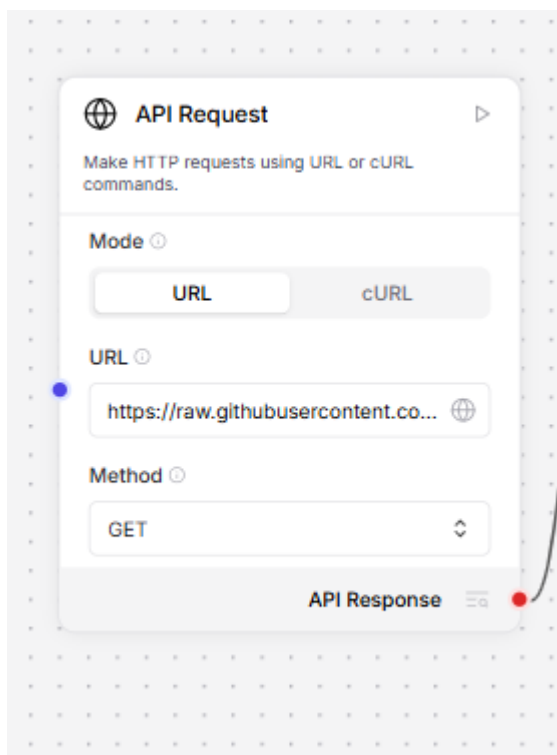


Task 3 : Code Reviewer Agent

Complete Flow



Step 1 : API Request



Component Output

Inspect the output of the component below.

Outputs Logs

```
text tree
{
  "source": "https://raw.githubusercontent.com/prathibaviswanathan/MyAssignments/refs/heads/main/week2/day1/LaunchBrowser.java",
  "status_code": 200,
  "response_headers": {
    "connection": "keep-alive",
    "content-length": "344",
    "cache-control": "max-age=300",
    "content-security-policy": "default-src 'none'; style-src 'unsafe-inline'; sandbox",
    "content-type": "text/plain; charset=utf-8",
    "etag": "W/\"e62c345eef4b8157a8fd73ad29156f136a1f4d83f304bd276b972e1aa34a4960\"",
    "strict-transport-security": "max-age=31536000",
    "x-content-type-options": "nosniff",
    "x-frame-options": "deny",
    "x-xss-protection": "1; mode=block",
    "x-github-request-id": "51D6:0A21:360864:72EEA1:69F774BC",
    "content-encoding": "gzip",
  }
}
```

Line: 1 Column: 1

Close

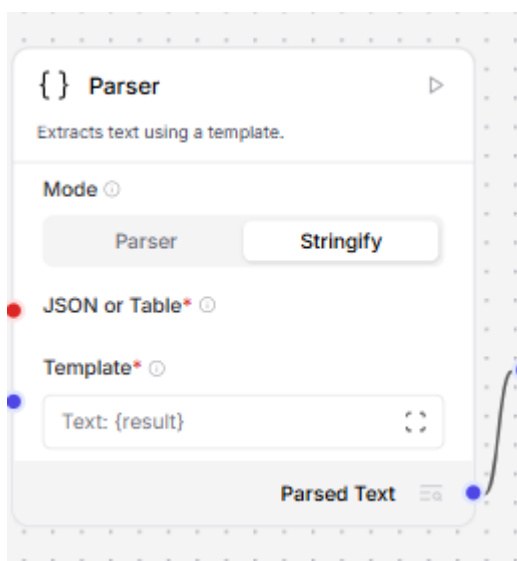
```
{
  "source":
"https://raw.githubusercontent.com/prathibaviswanathan/MyAssignments/refs/heads/main/we
ek2/day1/LaunchBrowser.java",
  "status_code": 200,
  "response_headers": {
    "connection": "keep-alive",
    "content-length": "344",
    "cache-control": "max-age=300",
    "content-security-policy": "default-src 'none'; style-src 'unsafe-inline'; sandbox",
    "content-type": "text/plain; charset=utf-8",
    "etag": "W/\"e62c345eef4b8157a8fd73ad29156f136a1f4d83f304bd276b972e1aa34a4960\"",
    "strict-transport-security": "max-age=31536000",
    "x-content-type-options": "nosniff",
    "x-frame-options": "deny",
    "x-xss-protection": "1; mode=block",
    "x-github-request-id": "51D6:0A21:360864:72EEA1:69F774BC",
    "content-encoding": "gzip",
    "accept-ranges": "bytes",
    "date": "Sun, 03 May 2026 17:09:03 GMT",
```

```

"via": "1.1 varnish",
"x-served-by": "cache-maa10238-MAA",
"x-cache": "HIT",
"x-cache-hits": "0",
"x-timer": "S1777828143.235244,VS0,VE226",
"vary": "Authorization,Accept-Encoding",
"access-control-allow-origin": "*",
"cross-origin-resource-policy": "cross-origin",
"x-fastly-request-id": "39d78922d80d7a9b4c00006d18b94a44ed1aa869",
"expires": "Sun, 03 May 2026 17:14:03 GMT",
"source-age": "0"
},
"result": "package week2.day1;\r\n\r\nimport
org.openqa.selenium.chrome.ChromeDriver;\r\n\r\npublic class LaunchBrowser
{\r\n\t\r\n\tpublic static void main(String[] args) {\r\n\t\t\r\n\t\tChromeDriver driver=new
ChromeDriver(); \r\n\t\t\r\n\t\tdriver.get(\"http://leafaps.com/opentaps/control/main\"); \r\n\t\t
driver.manage().window().maximize();\r\n\t\t
System.out.println(driver.getTitle());\r\n\t\t\r\n\t\t//System.setProperty(\"Webdriver.edhe.driver\
\", \"C:\\\\Users\\\\Viswa\\\\Desktop\\\\Prathiba\\\\Test Leaf
Software\\\\Edgedriver\");\r\n\t\t\r\n\t\t//EdgeDriver driver1=new
EdgeDriver();\r\n\t\t\r\n\t\t}\r\n\r\n}\r\n"
}

```

Step 2 : Parser



Component Output

Inspect the output of the component below.

Outputs

Logs

```
``json
{
  "source": "https://raw.githubusercontent.com/prathibaviswanathan/MyAssignments/refs/heads/main/week2/day1/LaunchBrowser.java",
  "status_code": 200,
  "response_headers": {
    "connection": "keep-alive",
    "content-length": "344",
    "cache-control": "max-age=300",
    "content-security-policy": "default-src 'none'; style-src 'unsafe-inline'; sandbox",
    "content-type": "text/plain; charset=utf-8",
    "etag": "W/\"e62c345eef4b8157a8fd73ad29156f136a1f4d83f304bd276b972e1aa34a4960\"",
    "strict-transport-security": "max-age=31536000",
    "x-content-type-options": "nosniff",
    "x-frame-options": "deny",
    "x-xss-protection": "1; mode=block",
    "x-github-request-id": "51D6:0A21:360864:72EEA1:69F774BC",
    "content-encoding": "gzip",
    "accept-ranges": "bytes",
    "date": "Sun, 03 May 2026 17:10:00 GMT",
```

Close

```
````json
{
 "source":
"https://raw.githubusercontent.com/prathibaviswanathan/MyAssignments/refs/heads/main/we
ek2/day1/LaunchBrowser.java",

 "status_code": 200,

 "response_headers": {

 "connection": "keep-alive",

 "content-length": "344",

 "cache-control": "max-age=300",

 "content-security-policy": "default-src 'none'; style-src 'unsafe-inline'; sandbox",

 "content-type": "text/plain; charset=utf-8",

 "etag": "W/\"e62c345eef4b8157a8fd73ad29156f136a1f4d83f304bd276b972e1aa34a4960\"",

 "strict-transport-security": "max-age=31536000",

 "x-content-type-options": "nosniff",

 "x-frame-options": "deny",

 "x-xss-protection": "1; mode=block",

 "x-github-request-id": "51D6:0A21:360864:72EEA1:69F774BC",

 "content-encoding": "gzip",

 "accept-ranges": "bytes",

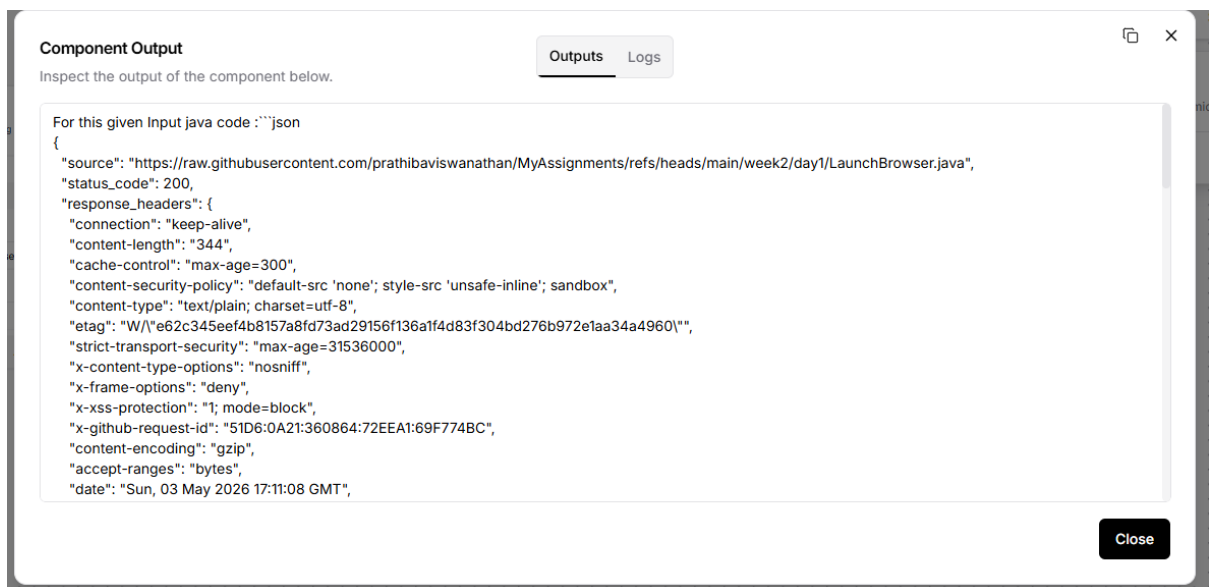
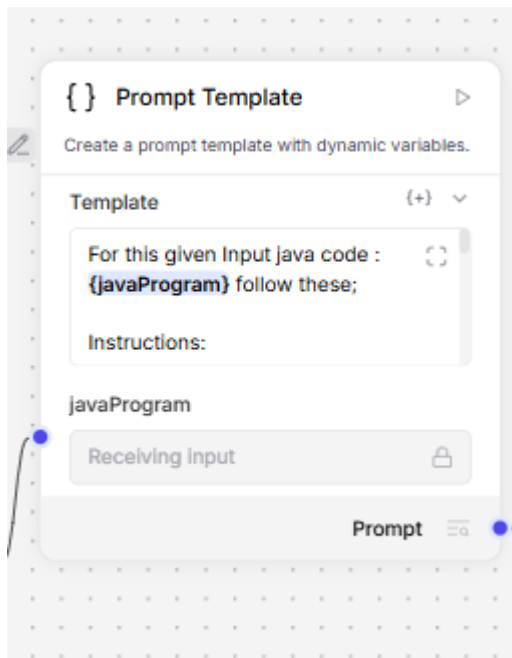
 "date": "Sun, 03 May 2026 17:10:00 GMT",
```

```

"via": "1.1 varnish",
"x-served-by": "cache-maa10233-MAA",
"x-cache": "HIT",
"x-cache-hits": "1",
"x-timer": "S1777828201.746907,VS0,VE1",
"vary": "Authorization,Accept-Encoding",
"access-control-allow-origin": "*",
"cross-origin-resource-policy": "cross-origin",
"x-fastly-request-id": "2f6cc682e74b132a02d9e26e51264362a983392a",
"expires": "Sun, 03 May 2026 17:15:00 GMT",
"source-age": "57"
},
"result": "package week2.day1;\r\n\r\nimport
org.openqa.selenium.chrome.ChromeDriver;\r\n\r\npublic class LaunchBrowser
{\r\n\t\r\n\tpublic static void main(String[] args) {\r\n\t\t\r\n\t\tChromeDriver driver=new
ChromeDriver(); \r\n\t\t\r\n\t\tdriver.get(\"http://leaptaps.com/opentaps/control/main\"); \r\n\t\t
driver.manage().window().maximize();\r\n\t\t
System.out.println(driver.getTitle());\r\n\t\t\r\n\t\t//System.setProperty(\"Webdriver.edhe.driver\",
\"C:\\\\Users\\\\Viswa\\\\Desktop\\\\Prathiba\\\\Test Leaf
Software\\\\Edgedriver\");\r\n\t\t\r\n\t\t//EdgeDriver driver1=new
EdgeDriver();\r\n\t\t\r\n\t\t}\r\n\r\n}
}
` ``

```

### Step 3 : Prompt



For this given Input java code : `` `json

{

"source":

"https://raw.githubusercontent.com/prathibaviswanathan/MyAssignments/refs/heads/main/week2/day1/LaunchBrowser.java",

"status\_code": 200,

"response\_headers": {

"connection": "keep-alive",

"content-length": "344",

```
"cache-control": "max-age=300",
"content-security-policy": "default-src 'none'; style-src 'unsafe-inline'; sandbox",
"content-type": "text/plain; charset=utf-8",
"etag": "W/\"e62c345eef4b8157a8fd73ad29156f136a1f4d83f304bd276b972e1aa34a4960\"",
"strict-transport-security": "max-age=31536000",
"x-content-type-options": "nosniff",
"x-frame-options": "deny",
"x-xss-protection": "1; mode=block",
"x-github-request-id": "51D6:0A21:360864:72EEA1:69F774BC",
"content-encoding": "gzip",
"accept-ranges": "bytes",
"date": "Sun, 03 May 2026 17:11:08 GMT",
"via": "1.1 varnish",
"x-served-by": "cache-maa10221-MAA",
"x-cache": "HIT",
"x-cache-hits": "0",
"x-timer": "S1777828268.296524,VS0,VE1",
"vary": "Authorization,Accept-Encoding",
"access-control-allow-origin": "*",
"cross-origin-resource-policy": "cross-origin",
"x-fastly-request-id": "a7e33d68e025aa39a369014254d69db0c17fbbd8",
"expires": "Sun, 03 May 2026 17:16:08 GMT",
"source-age": "125"
},
"result": "package week2.day1;\r\n\r\nimport
org.openqa.selenium.chrome.ChromeDriver;\r\n\r\npublic class LaunchBrowser
{\r\n\t\r\n\tpublic static void main(String[] args) {\r\n\t\t\r\n\t\tChromeDriver driver=new
ChromeDriver(); \r\n\t\t\r\n\t\tdriver.get(\"http://leaptaps.com/opentaps/control/main\"); \r\n\t\t
driver.manage().window().maximize();\r\n\t\t
System.out.println(driver.getTitle());\r\n\t\t\r\n\t\t//System.setProperty(\"Webdriver.edhe.driver\",
\"C:\\\\Users\\\\Viswa\\\\Desktop\\\\Prathiba\\\\Test Leaf
Software\\\\Edgedriver\");\r\n\t\t\r\n\t\t//EdgeDriver driver1=new
EdgeDriver();\r\n\t\t\r\n\t\t}\r\n\r\n}
}
```

``` follow these;

Instructions:

You are an expert Software Code Reviewer specialized in Java enterprise development and test automation frameworks.

Analyze the given Java code and provide:

Individual parameter scores (each out of 20)

Overall code quality score (out of 100)

Key improvement suggestions

Evaluate based on:

Code Readability

Code Efficiency

Error Handling

Maintainability

Best Practices & Standards

Provide output in this format:

Parameter | Description | Score (out of 20)

Code Readability | |

Code Efficiency | |

Error Handling | |

Maintainability | |

Best Practices & Standards | |

Overall Score: / 100

Improvement Areas:

<point 1>

<point 2>

<point 3>

Summary:

<2–3 lines summary>

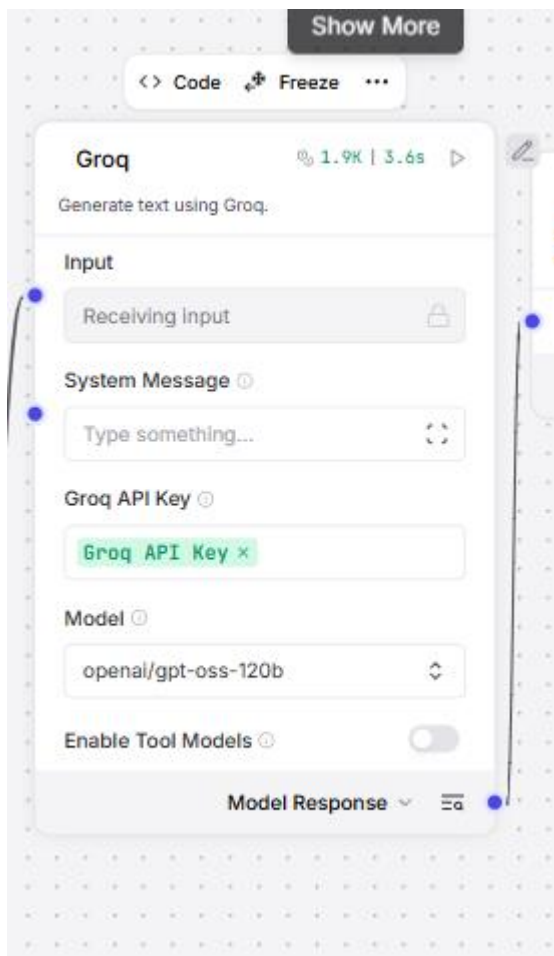
C – Context

The code is part of a production project and should meet clean code, maintainability, and basic quality standards.

P – Persona

You are a senior Java architect and code reviewer with strong experience in enterprise systems and automation frameworks.

Step 4 : LLM



Component Output

OutputsLogs

Inspect the output of the component below.

```
**Parameter	Description	Score (out of 20)**
**Code Readability** | The code is short and uses simple statements, but it lacks comments, contains hard-coded literals, and has a typo in a commented line. Package naming (`week2.day1`) is more tutorial-style than production-grade. | **12**
**Code Efficiency** | No obvious performance bottlenecks, but the driver is instantiated without any reuse strategy, and there are no implicit/explicit waits that could cause flaky runs. | **10**
**Error Handling** | No try-catch or finally blocks; any runtime exception (e.g., driver not found, navigation failure) will terminate the program abruptly. | **4**
**Maintainability** | Hard-coded URL and driver configuration, direct use of `ChromeDriver` instead of the `WebDriver` interface, and missing cleanup make future changes error-prone. | **8**
**Best Practices & Standards** | Violates several Selenium/Java best practices: missing `System.setProperty` (or `WebDriverManager`), no `driver.quit()`, no logging framework, no test framework (JUnit/TestNG), and inconsistent naming. | **6**

**Overall Score: 40 / 100**

---

### Improvement Areas
1. **Adopt proper WebDriver management** – use the `WebDriver` interface, set the driver binary via `System.setProperty` or a library like **WebDriverManager**, and ensure `driver.quit()` is called in a `finally` block or try-with-resources.
```

Close

****Parameter | Description | Score (out of 20)****

---|---|---

****Code Readability**** | The code is short and uses simple statements, but it lacks comments, contains hard-coded literals, and has a typo in a commented line. Package naming (`week2.day1`) is more tutorial-style than production-grade. | ****12****

****Code Efficiency**** | No obvious performance bottlenecks, but the driver is instantiated without any reuse strategy, and there are no implicit/explicit waits that could cause flaky runs. | ****10****

****Error Handling**** | No try-catch or finally blocks; any runtime exception (e.g., driver not found, navigation failure) will terminate the program abruptly. | ****4****

****Maintainability**** | Hard-coded URL and driver configuration, direct use of `ChromeDriver` instead of the `WebDriver` interface, and missing cleanup make future changes error-prone. | ****8****

****Best Practices & Standards**** | Violates several Selenium/Java best practices: missing `System.setProperty` (or `WebDriverManager`), no `driver.quit()`, no logging framework, no test framework (JUnit/TestNG), and inconsistent naming. | ****6****

****Overall Score: 40 / 100****

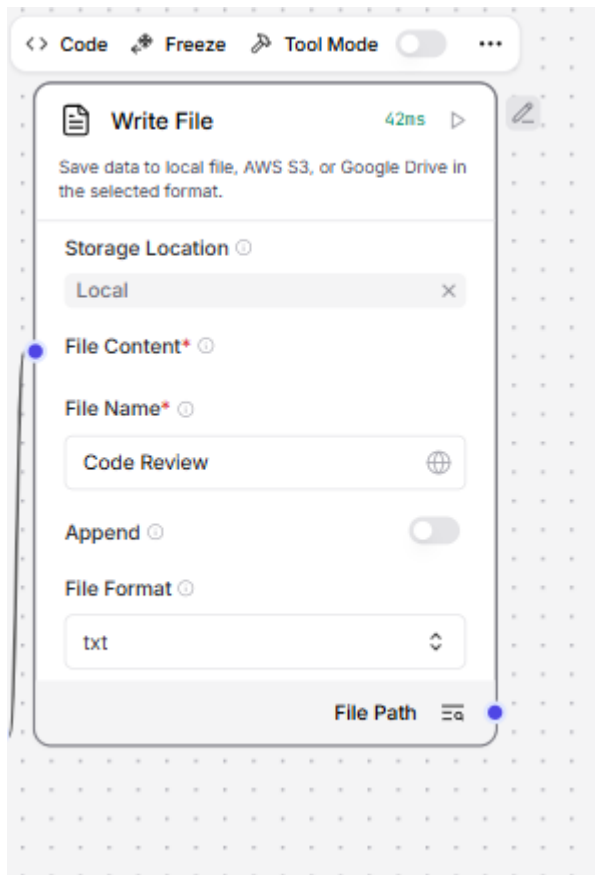
Improvement Areas

1. ****Adopt proper WebDriver management**** – use the `WebDriver` interface, set the driver binary via `System.setProperty` or a library like ****WebDriverManager****, and ensure `driver.quit()` is called in a `finally` block or `try-with-resources`.
2. ****Externalize configuration**** – move the application URL, driver paths, and other constants to a properties/YAML file or environment variables; avoid hard-coded strings.
3. ****Introduce robust error handling and logging**** – wrap Selenium actions in try-catch blocks, log meaningful messages with a logging framework (e.g., SLF4J), and fail gracefully rather than crashing the JVM.

****Summary****

The snippet demonstrates basic Selenium usage but falls short of production standards. Enhancing driver lifecycle management, externalizing configuration, and adding proper error handling and logging will markedly improve its quality, maintainability, and reliability.

Step 5 : Write File



Component Output

Inspect the output of the component below.

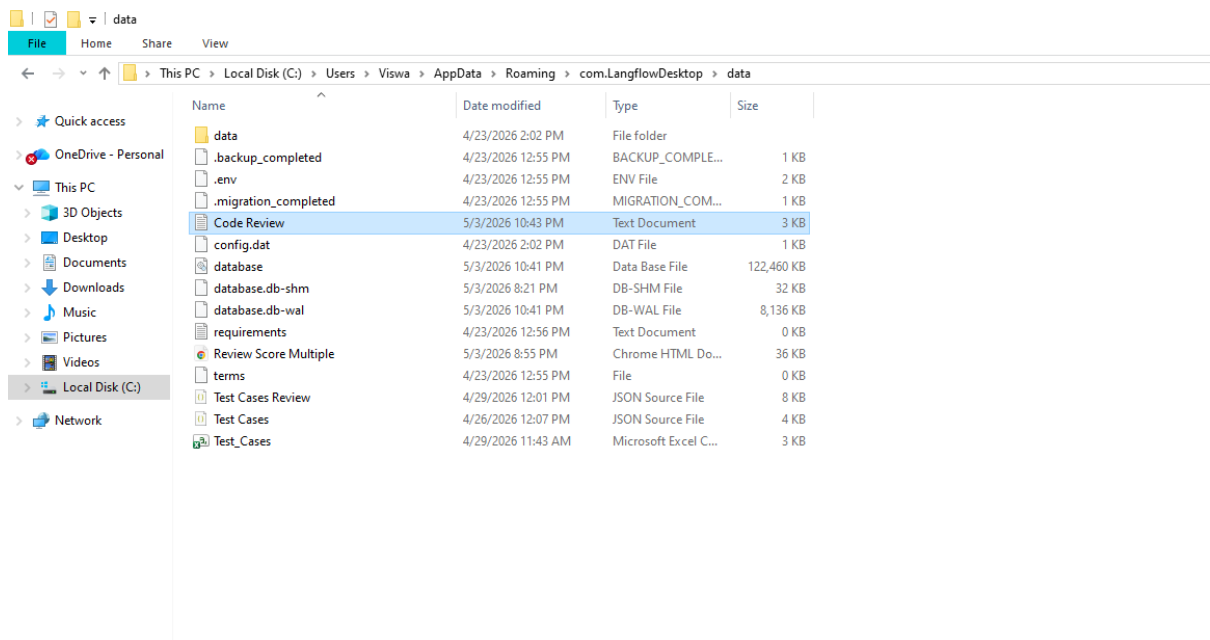
Outputs

Logs

✖

Message saved successfully as 'Code Review.txt' at C:\Users\Viswa\AppData\Roaming\com.LangflowDesktop\data\Code Review.txt

Close



Output

****Parameter | Description | Score (out of 20)****

---|---|---

****Code Readability**** | The code is short and indented, but it lacks comments, uses magic strings (hard-coded URL), and mixes production code with commented-out experimental code. Variable naming is acceptable but could be more expressive. | ****12****

****Code Efficiency**** | The snippet only launches a browser; there is no measurable inefficiency, but it also does not reuse a driver instance or apply any wait strategies. | ****10****

****Error Handling**** | No try-catch blocks, no validation of driver creation, and no cleanup (`` quit() ``). Any runtime exception will terminate the JVM abruptly. | ****4****

****Maintainability**** | Hard-coded values, direct use of `` ChromeDriver `` instead of the `` WebDriver `` interface, and absence of configuration files make future changes error-prone. The commented Edge driver code adds noise. | ****8****

****Best Practices & Standards**** | Violates several Selenium and Java conventions: missing `` System.setProperty `` (or `` WebDriverManager ``) for driver binaries, no use of a logging framework, no `` finally `` block for resource release, and the package name (`` week2.day1 ``) is not production-grade. | ****6****

****Overall Score: 40 / 100****

Improvement Areas

1. ****Driver Setup & Cleanup****

- Use ``WebDriverManager`` or ``System.setProperty`` to locate the driver binary.
- Declare the driver as ``WebDriver driver = new ChromeDriver();`` and ensure ``driver.quit()`` is called in a ``finally`` block or via a try-with-resources pattern.

2. ****Configuration & Maintainability****

- Externalize the URL (and any other environment-specific values) to a properties/YAML file or use environment variables.
- Remove dead/commented code; keep the source clean.

3. ****Error Handling & Logging****

- Wrap Selenium actions in appropriate try-catch blocks, log exceptions using a framework like SLF4J/Log4j, and provide meaningful messages.
- Consider using a test framework (TestNG/JUnit) instead of a raw ``main`` method for better reporting and reusability.

****Summary:****

The current snippet demonstrates a basic Selenium launch but falls short of production standards due to missing driver configuration, lack of error handling, and hard-coded values. Refactoring to use proper WebDriver setup, external configuration, robust cleanup, and logging will markedly improve its quality and maintainability.